

## 6.4.3 Numerical stability of linear solve via LU factorization



6.4.3 Backward error for solution of linear system

fit width



Let us now combine the results from [Subsection 6.4.1](#) and [Subsection 6.4.2](#) into a backward error result for solving  $Ax = y$  via LU factorization and two triangular solves.

**Theorem 6.4.3.1.** *Let  $A \in \mathbb{R}^{n \times n}$  and  $x, y \in \mathbb{R}^n$  with  $Ax = y$ . Let  $\tilde{x}$  be the approximate solution computed via the following steps:*

- *Compute the LU factorization, yielding approximate factors  $\tilde{L}$  and  $\tilde{U}$ .*
- *Solve  $\tilde{L}z = y$ , yielding approximate solution  $\tilde{z}$ .*
- *Solve  $\tilde{U}\tilde{x} = \tilde{z}$ , yielding approximate solution  $\tilde{x}$ .*

Then

$$(A + \Delta A)\tilde{x} = y \quad \text{with} \quad |\Delta A| \leq (3\gamma_n + \gamma_n^2)|\tilde{L}||\tilde{U}|.$$

We refer the interested learner to the proof in the previously mentioned papers [\[6\]](#) [\[7\]](#).

**Homework 6.4.3.1.** The question left is how a change in a nonsingular matrix affects the accuracy of the solution of a linear system that involves that matrix. We saw in [Subsection 1.4.1](#) that if

$$Ax = y \text{ and } A(x + \delta x) = y + \delta y$$

then

$$\frac{\|\delta x\|}{\|x\|} \leq \kappa(A) \frac{\|\delta y\|}{\|y\|}$$

when  $\|\cdot\|$  is a subordinate norm. But what we want to know is how a change in  $A$  affects the solution:

$$Ax = y \text{ and } (A + \Delta A)(x + \delta x) = y$$

then

$$\frac{\|\delta x\|}{\|x\|} \leq \frac{\kappa(A) \frac{\|\Delta A\|}{\|A\|}}{1 - \kappa(A) \frac{\|\Delta A\|}{\|A\|}}.$$

Prove this!

▼ [Solution](#)

$$Ax = y \text{ and } (A + \Delta A)(x + \delta x) = y$$

implies that

$$(A + \Delta A)(x + \delta x) = Ax$$

or, equivalently,

$$\Delta Ax + A\delta x + \Delta A\delta x = 0.$$

We can rewrite this as

$$\delta x = A^{-1}(-\Delta Ax - \Delta A\delta x)$$

so that

$$\|\delta x\| = \|A^{-1}(-\Delta Ax - \Delta A\delta x)\| \leq \|A^{-1}\| \|\Delta A\| \|x\| + \|A^{-1}\| \|\Delta A\| \|\delta x\|.$$

This can be rewritten as

$$(1 - \|A^{-1}\| \|\Delta A\|) \|\delta x\| \leq \|A^{-1}\| \|\Delta A\| \|x\|$$

and finally

$$\frac{\|\delta x\|}{\|x\|} \leq \frac{\|A^{-1}\| \|\Delta A\|}{1 - \|A^{-1}\| \|\Delta A\|}$$

and finally

$$\frac{\|\Delta x\|}{\|x\|} \leq \frac{\|A\| \|A^{-1}\| \frac{\|\Delta A\|}{\|A\|}}{1 - \|A\| \|A^{-1}\| \frac{\|\Delta A\|}{\|A\|}}.$$

The last homework brings up a good question: If  $A$  is nonsingular, how small does  $\Delta A$  need to be for it to be nonsingular?

**Theorem 6.4.3.2.** *Let  $A$  be nonsingular,  $\|\cdot\|$  be a subordinate norm, and*

$$\frac{\|\Delta A\|}{\|A\|} < \frac{1}{\kappa(A)}.$$

*Then  $A + \Delta A$  is nonsingular.*

*Proof.*

Proof by contradiction.

Assume that  $A$  is *nonsingular*,

$$\frac{\|\Delta A\|}{\|A\|} < \frac{1}{\kappa(A)}. \quad (6.4.5)$$

and  $A + \Delta A$  is singular. We will show this leads to a contradiction.

Since  $A + \Delta A$  is singular, there exists  $x \neq 0$  such that  $(A + \Delta A)x = 0$ . We can rewrite this as

$$x = -A^{-1}\Delta Ax$$

and hence

$$\|x\| = \|A^{-1}\Delta Ax\| \leq \|A^{-1}\| \|\Delta A\| \|x\|.$$

Dividing both sides by  $\|x\|$  yields

$$1 \leq \|A^{-1}\| \|\Delta A\|$$

and hence  $\frac{1}{\|A^{-1}\|} \leq \|\Delta A\|$  and finally

$$\frac{1}{\|A\| \|A^{-1}\|} \leq \frac{\|\Delta A\|}{\|A\|},$$

which contradicts (6.4.5) since  $\|A\| \|A^{-1}\| = \kappa(A)$ .